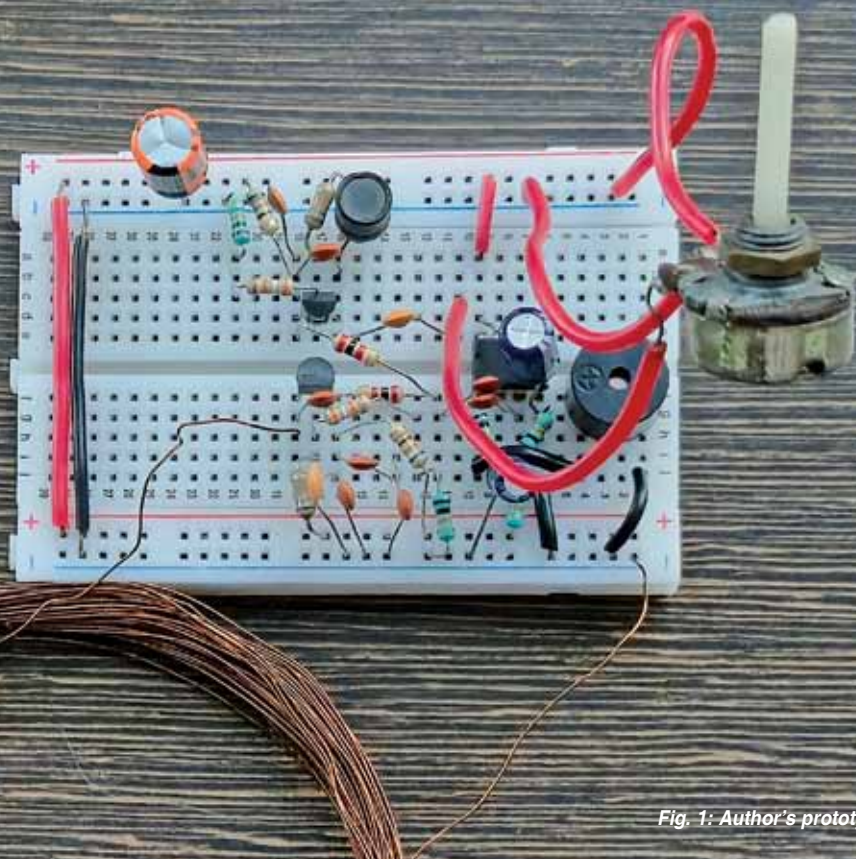
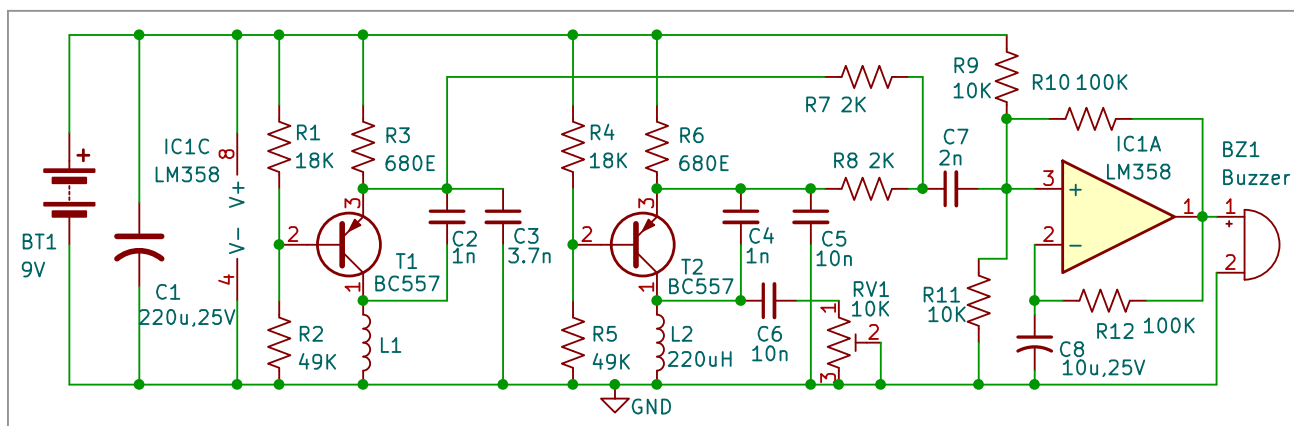


ABHAY VERMA

This BFO (beat frequency oscillator) metal detector system demonstrates the practical application of analogue signal mixing and processing techniques. Beat frequency oscillator provides one of the simplest and most reliable methods for detecting nearby metallic objects. By combining classical radio-frequency oscillator techniques with a modern

The circuit consists of two LC oscillators built using BC557 transistors, inductors, and capacitors. The search oscillator employs a 30-turn search coil wound with 30AWG enamelled copper wire on a former of approximately 10-centimetre diameter. This oscillator operates at

The outputs of these two high-frequency oscillators are then mixed. Since their operating frequencies are nearly identical, the resulting



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